Spatial Autocorrelation: Global and Local Moran's I

Do high SIRs sit next to high SIRs?

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Get the source. Download `index.qmd` — open it in RStudio and run the chunks alongside this page.

Where we left off

Last time we turned Pennsylvania's lung cancer cases into expected counts and SIRs. Looking at the SIR map, the eye starts to see patterns: a darker patch here, a lighter one there. Today we ask whether those patterns are real, in two steps.

1. **Global Moran's I** — one number for the whole state: do similar SIRs tend to sit next to each other?
2. **Local Moran's I (LISA)** — one number per county: *where* are the clusters and the outliers?

Between the two, we use **simulation** to decide what "more clustered than chance" means. Reading: **Rogerson & Yamada Ch. 3–4**.

```
library(sf)
library(dplyr)
library(ggplot2)
library(patchwork)
library(spdep)        # neighbours, weights, Moran's I
library(SpatialEpi)   # pennLC data

theme_set(theme_classic(base_size = 12))
```

> **Note**
>
> If `spdep` is missing, `install.packages("spdep")` once.

Back to the SIR map

We rebuild the observed and expected counts from last lecture in one chunk.

```
data(pennLC)
d <- pennLC$data

ref <- d |>
  group_by(race, gender, age) |>
  summarise(r = sum(cases) / sum(population), .groups = "drop")   # 16 reference rates

county <- d |>
  left_join(ref, by = c("race", "gender", "age")) |>
  group_by(county) |>
  summarise(O = sum(cases), E = sum(population * r), .groups = "drop") |>
  mutate(SIR = O / E)

pa <- st_as_sf(pennLC$spatial.polygon) |>
  st_set_crs(4326) |>
  mutate(county = names(pennLC$spatial.polygon)) |>
  st_transform(5070) |>                                          # Albers, metres
  left_join(county, by = "county")

sir_breaks <- c(0, 0.67, 0.83, 0.95, 1.05, 1.2, 1.5, Inf)
sir_fill   <- scale_fill_brewer(palette = "RdBu", direction = -1, drop = FALSE, name = "SIR")
```

```
ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill +
  theme_void()
```

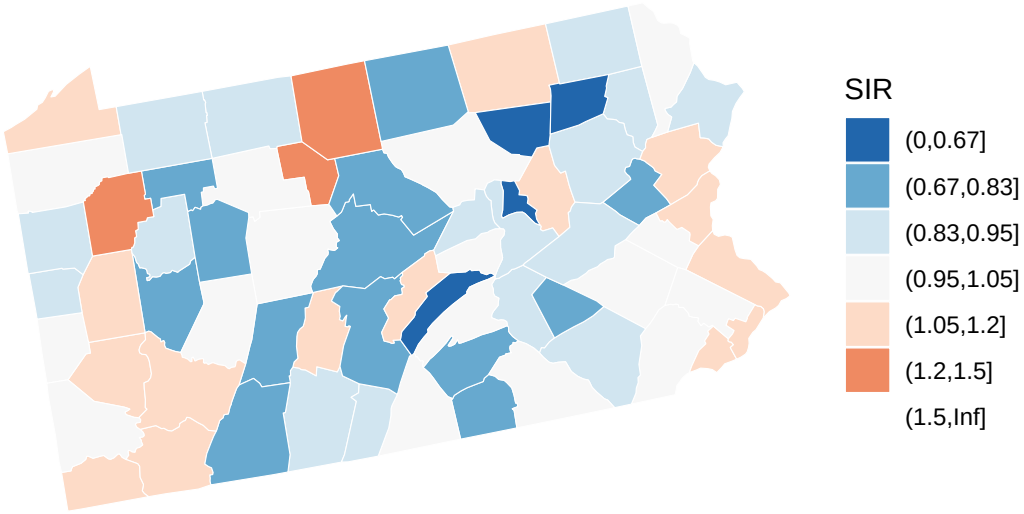


Figure 1: Raw SIRs for lung cancer, Pennsylvania 2002.

Our eyes are good at finding patterns, including ones that are not there. We need (a) a definition of "nearby" and (b) a measure of how similar nearby values are.

Who is a neighbour?

Queen contiguity

Two counties are neighbours if they share a border or even a single corner (like the queen in chess). `poly2nb()` builds the list.

```
nb <- poly2nb(pa, queen = TRUE)
summary(card(nb))                                  # neighbours per county
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.000   4.000   5.000   5.164   6.000   9.000
```

```r
pts   <- st_point_on_surface(st_geometry(pa))
links <- nb2lines(nb, coords = pts, as_sf = TRUE) |>
  st_set_crs(st_crs(pa))

i_leh <- which(pa$county == "lehigh")
pa$role <- "other"
pa$role[nb[[i_leh]]] <- "neighbour"
pa$role[i_leh] <- "Lehigh"

ggplot() +
  geom_sf(data = pa, aes(fill = role), colour = "white", linewidth = 0.3) +
  geom_sf(data = links, colour = "grey30", linewidth = 0.25) +
  geom_sf(data = pts, size = 0.6) +
  scale_fill_manual(values = c(Lehigh = "#b2182b", neighbour = "#fdae61", other = "grey90"),
                    name = NULL) +
  theme_void()
```

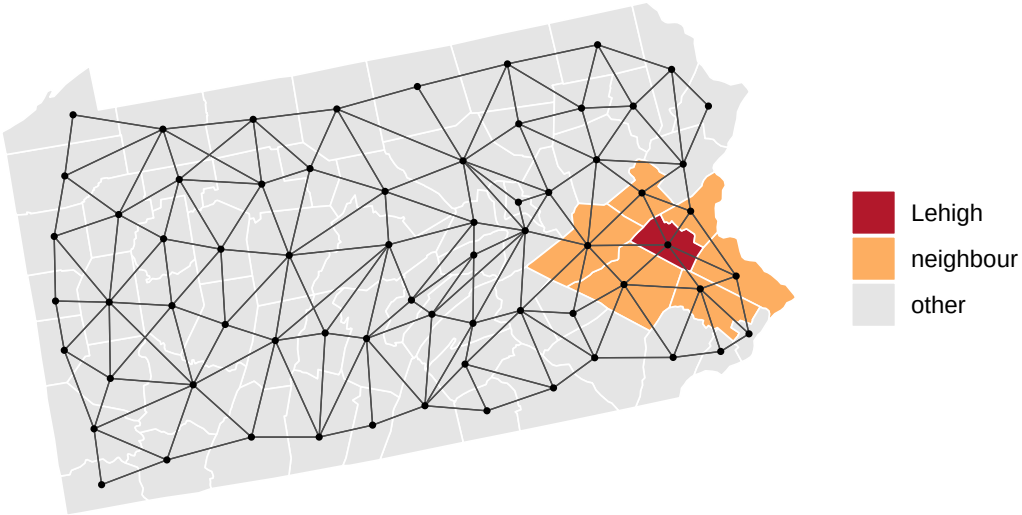


Figure 2: Queen-contiguity links between county centres. Lehigh and its neighbours are highlighted.

Other choices exist — rook contiguity (shared edges only), the k nearest centroids, or a distance band — and the answer can depend on the choice. Contiguity is the usual default for counties.

Row-standardized weights

We turn the list into weights. With `style = "W"`, each county's neighbours get equal weight and the weights sum to 1, so "the neighbours' value" is simply their **average**.

```
lw <- nb2listw(nb, style = "W")

lw$weights[[i_leh]]                              # Lehigh: equal shares
```

```
[1] 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667 0.1666667
```

The spatial lag

The **spatial lag** of a county is the average SIR of its neighbours. Mapping it next to the SIR is the most intuitive way to see autocorrelation: if the two maps look alike, neighbours resemble each other.

```
pa$SIR_lag <- lag.listw(lw, pa$SIR)

m_own <- ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill + labs(title = "Own SIR")

m_lag <- ggplot(pa) +
  geom_sf(aes(fill = cut(SIR_lag, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill + labs(title = "Neighbours' average")

(m_own | m_lag) + plot_layout(guides = "collect") & theme_void()
```

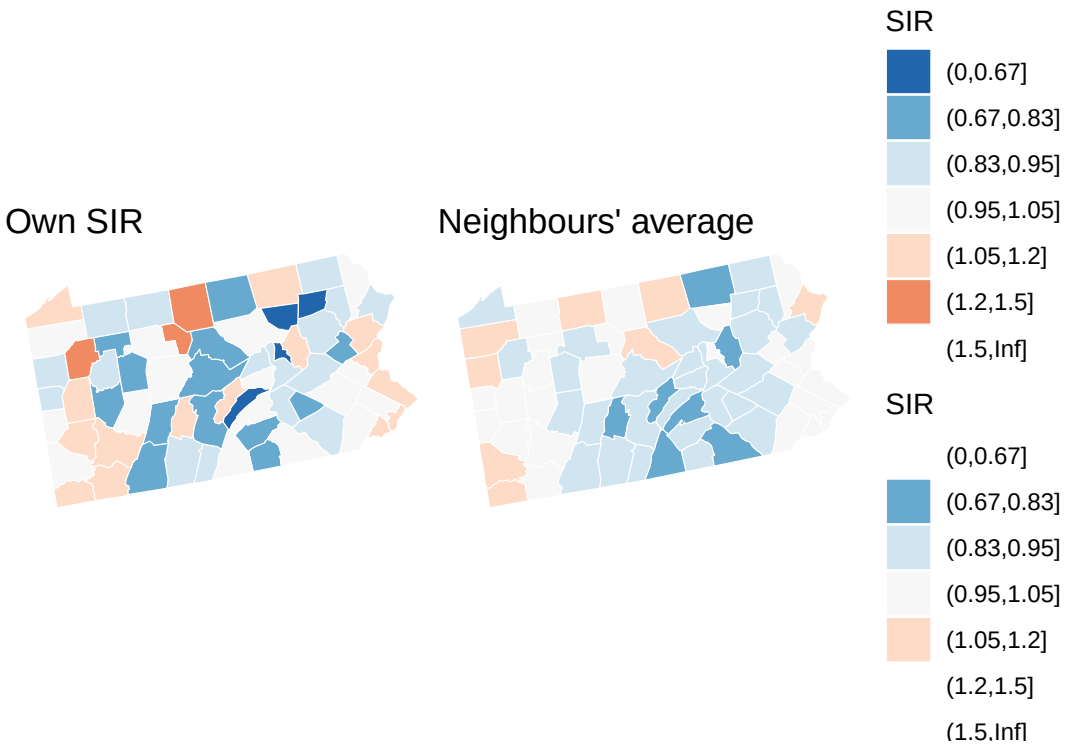


Figure 3: Each county's own SIR (left) and the average SIR of its neighbours (right).

The lag map is smoother, as averages always are. The question is whether high-SIR counties tend to have high-SIR neighbours.

Global Moran's I

The Moran scatterplot

Plot each county's (standardized) SIR against its neighbours' average.

```
pa <- pa |>
  mutate(z     = as.numeric(scale(SIR)),        # standardized SIR
         z_lag = lag.listw(lw, z))              # neighbours' average

ggplot(pa, aes(z, z_lag)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_point(aes(size = E), alpha = 0.5, colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, colour = "#b2182b", linewidth = 0.8) +
  geom_text(data = \(x) slice_max(x, abs(z * z_lag), n = 5),
```

```
          aes(label = county), size = 3, vjust = -0.8) +
annotate("label", x = Inf, y = Inf, label = "High-High", hjust = 1.1, vjust = 1.3, colour =
annotate("label", x = -Inf, y = -Inf, label = "Low-Low", hjust = -0.1, vjust = -0.3, colour =
annotate("label", x = Inf, y = -Inf, label = "High-Low", hjust = 1.1, vjust = -0.3, colour =
annotate("label", x = -Inf, y = Inf, label = "Low-High", hjust = -0.1, vjust = 1.3, colour =
labs(x = "SIR (standardized)", y = "Neighbours' average (standardized)", size = "Expected")
```

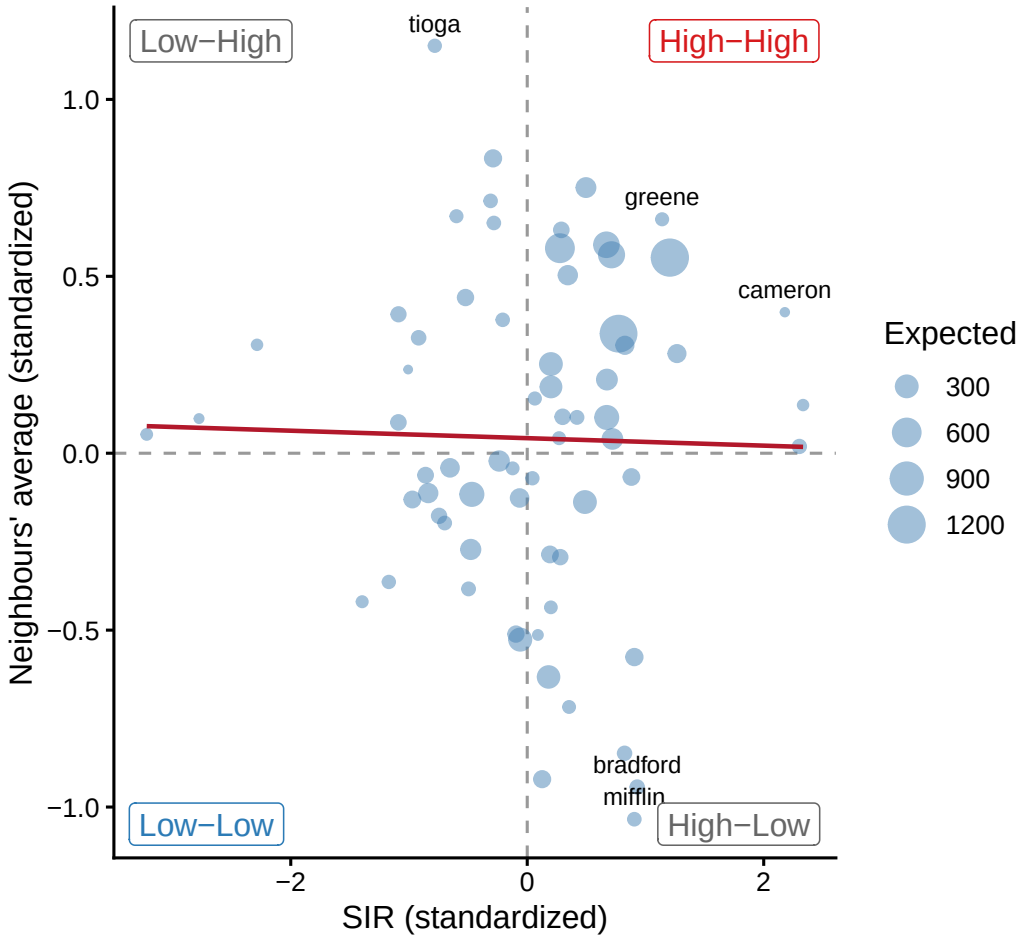


Figure 4: Moran scatterplot. The slope of the line is Moran's I.

Points in the **High-High** and **Low-Low** quadrants push the slope up (similar neighbours); points in **High-Low** and **Low-High** pull it down (dissimilar neighbours). With row-standardized weights, the slope of this line *is* Moran's I.

The statistic

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}\,(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

Read it as a correlation between each value and its neighbours' values. Values near 0 suggest no pattern (strictly, the null expectation is $-1/(n-1)$, about -0.015 for 67 counties); positive values mean similar values cluster; negative values mean a checkerboard, which is rare for disease data.

```
mi <- moran.test(pa$SIR, lw)

c(slope     = unname(coef(lm(z_lag ~ z, data = pa))[2]),
  Moran_I   = unname(mi$estimate[1]),
  expected  = unname(mi$estimate[2]))
```

```
      slope     Moran_I    expected
-0.01055968 -0.01055968 -0.01515152
```

The two numbers agree: Moran's I for lung cancer SIRs is -0.011. Is that large? On its own, we cannot tell. That is what simulation is for.

Is it more than chance? Simulation

Shuffle the map

The null hypothesis says location does not matter: any county could have had any SIR. So we **shuffle** the 67 SIRs at random across the counties and look at the result.

```
set.seed(374)
shuffled <- bind_rows(
  pa |> mutate(panel = "Observed"),
  pa |> mutate(SIR = sample(SIR), panel = "Shuffle 1"),
  pa |> mutate(SIR = sample(SIR), panel = "Shuffle 2"),
  pa |> mutate(SIR = sample(SIR), panel = "Shuffle 3")
)

ggplot(shuffled) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.1) +
  sir_fill +
  facet_wrap(~ panel) +
  theme_void()
```

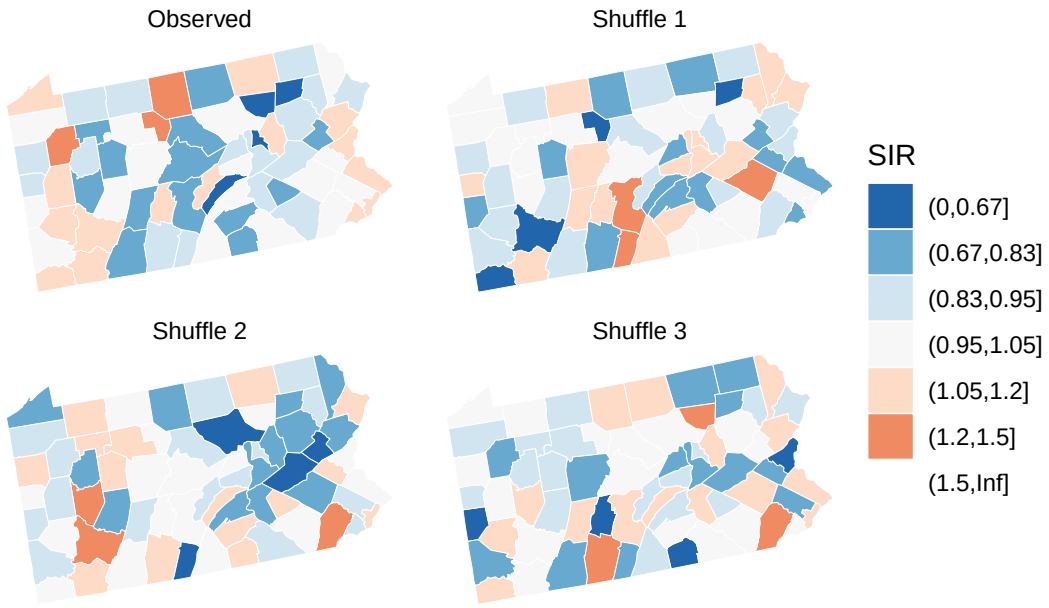


Figure 5: The real SIR map (top left) and three random shufflings of the same 67 values.

Same values, same colours, different arrangement. Can you pick out the real map? If not easily, the pattern may be weak. Moran's I turns this eyeball test into a number.

Do it 999 times

`moran.mc()` shuffles the values 999 times, computes I each time, and compares the observed I with the shuffled ones.

```
set.seed(374)
mc <- moran.mc(pa$SIR, lw, nsim = 999)
mc
```

```
    Monte-Carlo simulation of Moran I

data:  pa$SIR
weights: lw
number of simulations + 1: 1000

statistic = -0.01056, observed rank = 552, p-value = 0.448
alternative hypothesis: greater
```

```r
sims <- data.frame(I = head(mc$res, -1))        # last element is the observed I

ggplot(sims, aes(I)) +
  geom_histogram(bins = 40, fill = "grey75", colour = "white") +
  geom_vline(xintercept = mc$statistic, colour = "#b2182b", linewidth = 1) +
  geom_vline(xintercept = -1 / (nrow(pa) - 1), linetype = "dashed") +
  labs(x = "Moran's I", y = "Number of shuffles")
```

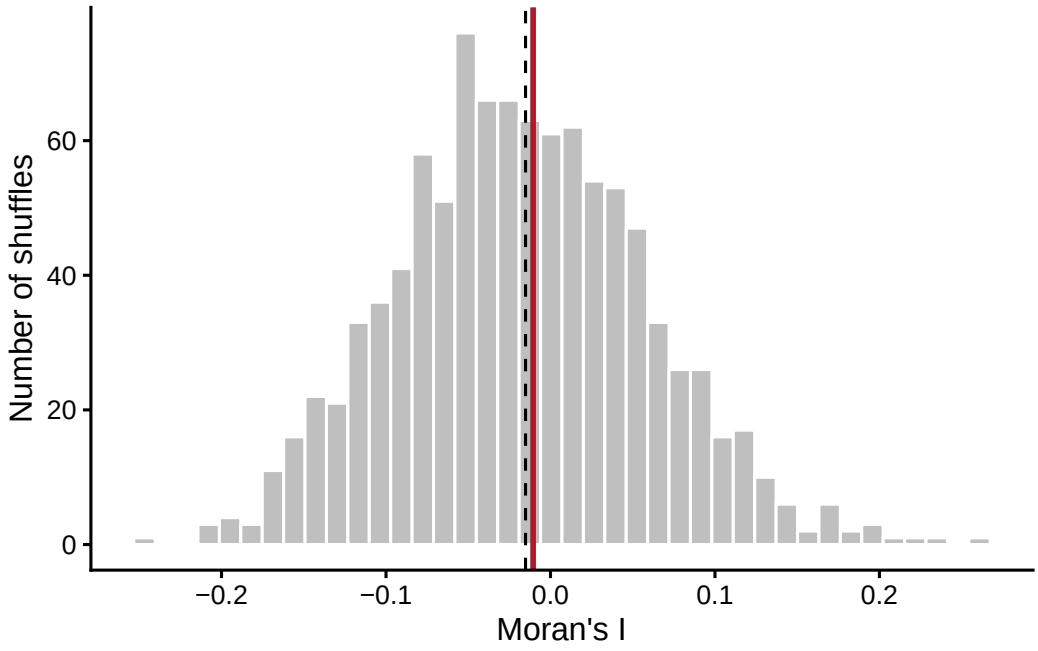


Figure 6: Moran's I under 999 random shufflings (grey) and the observed value (red).

The pseudo p-value counts how many shuffles did at least as well as the real map:

$$p = \frac{R + 1}{999 + 1},$$

where R is the number of shuffled I's at or above the observed one. Here $p = 0.45$. With 999 shuffles, the smallest possible value is 0.001; more shuffles give finer resolution, not stronger evidence.

! SIRs are not equally reliable

The shuffle treats Forest County (a handful of expected cases) and Philadelphia (over a thousand) as interchangeable. Small counties produce extreme SIRs by chance, and those extremes can create or hide apparent clustering. An Empirical Bayes version of Moran's I (`spdep::EBImoran.mc()`, Assunção & Reis 1999) adjusts for this; we return to it with smoothing.

Local Moran's I (LISA)

From one number to 67

Global I says *whether* the map is clustered, not *where*. The local version gives each county its own value:

$$I_i \propto (x_i - \bar{x}) \times \text{average of } (x_j - \bar{x}) \text{ over neighbours } j.$$

Positive I_i: the county resembles its neighbours (High-High or Low-Low). Negative I_i: it differs from them (High-Low or Low-High). The local values average to the global I — each county's share of the overall pattern.

Conditional permutation

Significance is again simulated, with one twist: county i is **held fixed** and only the other values are shuffled around it. This asks whether i's particular neighbourhood is unusual, given i's own value.

```
set.seed(374)
lisa <- localmoran_perm(pa$SIR, lw, nsim = 999)

pa <- pa |>
  mutate(
    Ii    = lisa[, "Ii"],
    p_sim = lisa[, "Pr(z != E(Ii)) Sim"],                    # simulation-based p
    quad  = as.character(attr(lisa, "quadr")$mean)           # scatterplot quadrant
  )

c(mean_local = mean(pa$Ii), global = unname(mi$estimate[1]))  # same number
```

```
 mean_local      global
-0.01055968 -0.01055968
```

```r
m_Ii <- ggplot(pa) +
  geom_sf(aes(fill = Ii), colour = "white", linewidth = 0.2) +
  scale_fill_gradient2(low = "#2166ac", mid = "grey95", high = "#b2182b", name = "Local I") +
  labs(title = "Local I")

m_p <- ggplot(pa) +
  geom_sf(aes(fill = cut(p_sim, c(0, 0.01, 0.05, 0.1, 1))), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = c("#7f0000", "#d7301f", "#fdbb84", "grey95"),
                    labels = c("< 0.01", "0.01--0.05", "0.05--0.10", "> 0.10"),
                    drop = FALSE, name = "p") +
  labs(title = "Significance")

(m_Ii | m_p) & theme_void()
```

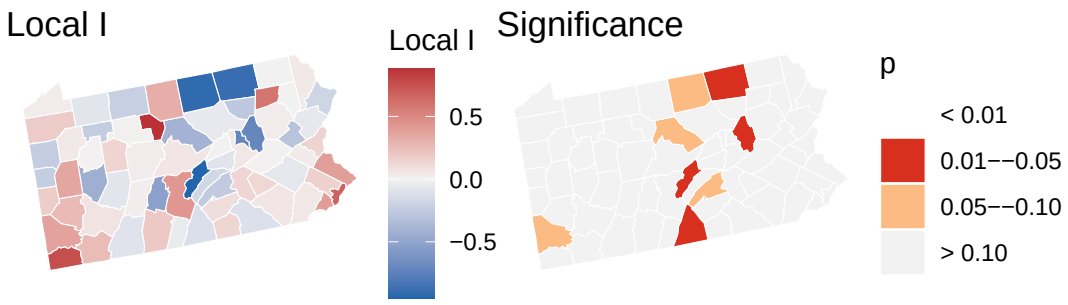


Figure 7: Local Moran's I (left) and its simulated p-value (right).

The LISA cluster map

Combining the two maps — quadrant for significant counties, grey otherwise — gives the classic LISA map.

```r
lisa_cols <- c("High-High" = "#d7191c", "Low-Low" = "#2c7bb6",
               "High-Low"  = "#fdae61", "Low-High" = "#abd9e9",
               "Not significant" = "grey92")
```

```r
pa <- pa |>
  mutate(cluster = factor(if_else(p_sim < 0.05, quad, "Not significant"),
                          levels = names(lisa_cols)))

m_lisa <- ggplot(pa) +
  geom_sf(aes(fill = cluster), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = lisa_cols, drop = FALSE, name = NULL) +
  labs(title = "LISA clusters")

(m_own | m_lisa) & theme_void()
```

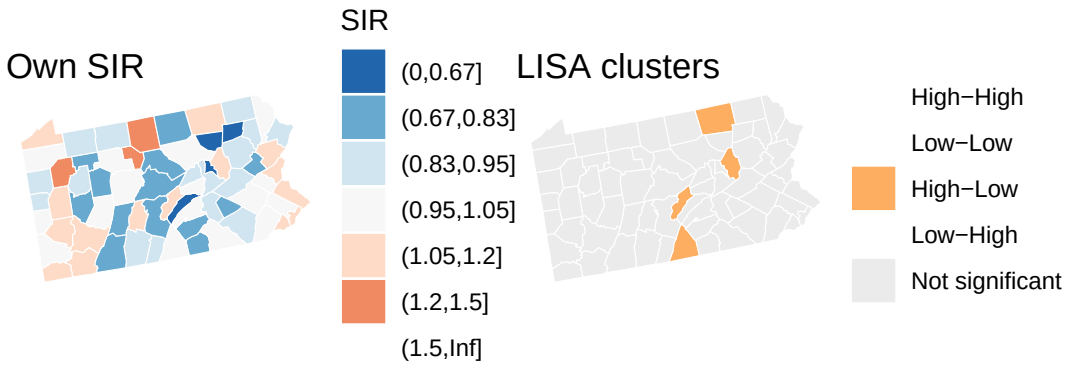


Figure 8: SIR (left) and LISA clusters at p < 0.05 (right).

```r
pa |>
  st_drop_geometry() |>
  filter(cluster != "Not significant") |>
  arrange(cluster, p_sim) |>
  transmute(county, cluster, O, E = round(E, 1), SIR = round(SIR, 2),
            Ii = round(Ii, 2), p_sim)
```

```
    county  cluster   O     E  SIR    Ii p_sim
1 franklin High-Low 105 109.9 0.96 -0.12 0.030
2 columbia High-Low  58  53.3 1.09 -0.71 0.032
```

```
3  mifflin High-Low  45  40.8 1.10 -0.95 0.042
4 bradford High-Low  59  53.2 1.11 -0.89 0.048
```

- **High-High** (hot spot): high SIR, high-SIR neighbours.
- **Low-Low** (cold spot): low SIR, low-SIR neighbours.
- **High-Low / Low-High** (spatial outliers): a county that breaks from its surroundings.

A High-High county is the *centre* of a cluster; its neighbours belong to the cluster but need not be significant themselves.

How much should we believe it?

```
m_strict <- pa |>
  mutate(cluster = factor(if_else(p_sim < 0.01, quad, "Not significant"),
                          levels = names(lisa_cols))) |>
  ggplot() +
  geom_sf(aes(fill = cluster), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = lisa_cols, drop = FALSE, name = NULL) +
  labs(title = "p < 0.01")

(m_lisa + labs(title = "p < 0.05") | m_strict) +
  plot_layout(guides = "collect") & theme_void()
```

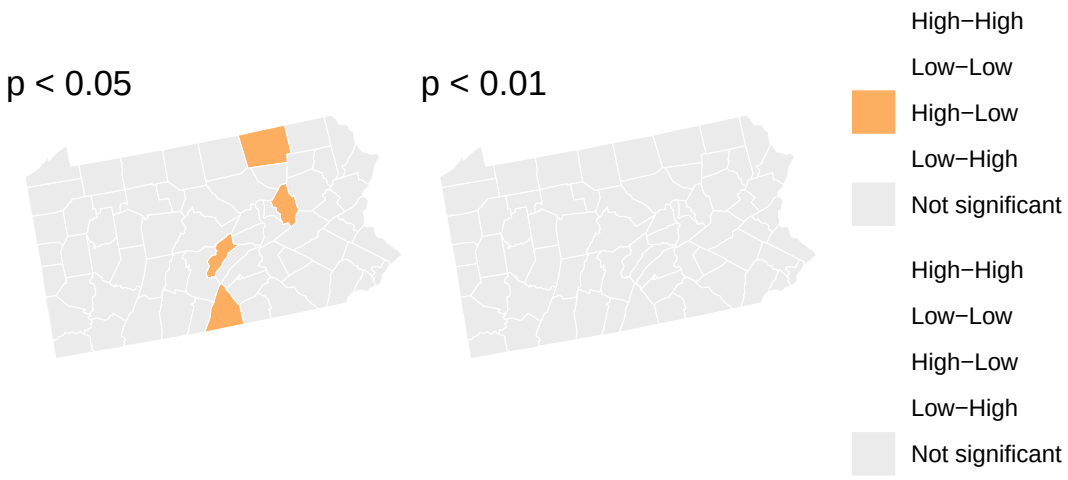


Figure 9: LISA clusters at p $<$ 0.05 (left) and p $<$ 0.01 (right).

⚠ 67 tests, not one

At $p < 0.05$, about $0.05 \times 67 \approx 3$ counties would be flagged even if lung cancer risk had no spatial structure at all. Neighbouring tests also share data, so they are not independent. Treat LISA output as a map of places worth a closer look, not as a list of confirmed clusters. Clusters that survive a stricter threshold, involve counties with large E_i, and make epidemiological sense deserve the most attention.

At home exercise: fifteen minutes

1. Replace queen contiguity with the four nearest neighbours (`knn2nb(knearneigh(st_coordinates(pts), k = 4))`). Does global I change much? Does the LISA map?
2. In the LISA table, which flagged counties have $E_i < 20$? Would you report them to a county health officer? Answer in one sentence.
3. Explain, in one sentence for a non-specialist, what a High-Low outlier means.

Key takeaways

1. Spatial autocorrelation needs a definition of neighbours; with row-standardized weights, the spatial lag is the neighbours' average.
2. Global Moran's I is the slope of the Moran scatterplot: one number for the whole map.
3. Permutation answers "compared to what?": shuffle the values, recompute, and see where the observed I falls.
4. Local Moran's I locates hot spots, cold spots, and outliers, using conditional permutation for each county.
5. Raw SIRs differ in reliability, and LISA runs many tests at once; read both statistics with that in mind.

References

Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2), 93–115.

Assunção, R. M., & Reis, E. A. (1999). A new proposal to adjust Moran's I for population density. *Statistics in Medicine*, 18(16), 2147–2162.

Bivand, R. S., & Wong, D. W. S. (2018). Comparing implementations of global and local indicators of spatial association. *TEST*, 27(3), 716–748.

Cliff, A. D., & Ord, J. K. (1981). *Spatial processes: Models and applications*. Pion.

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23.

Rogerson, P., & Yamada, I. (2009). *Statistical detection and surveillance of geographic clusters*. Chapman & Hall/CRC, Ch. 3–4.